

HW2 – Part 1

Preparation - 1

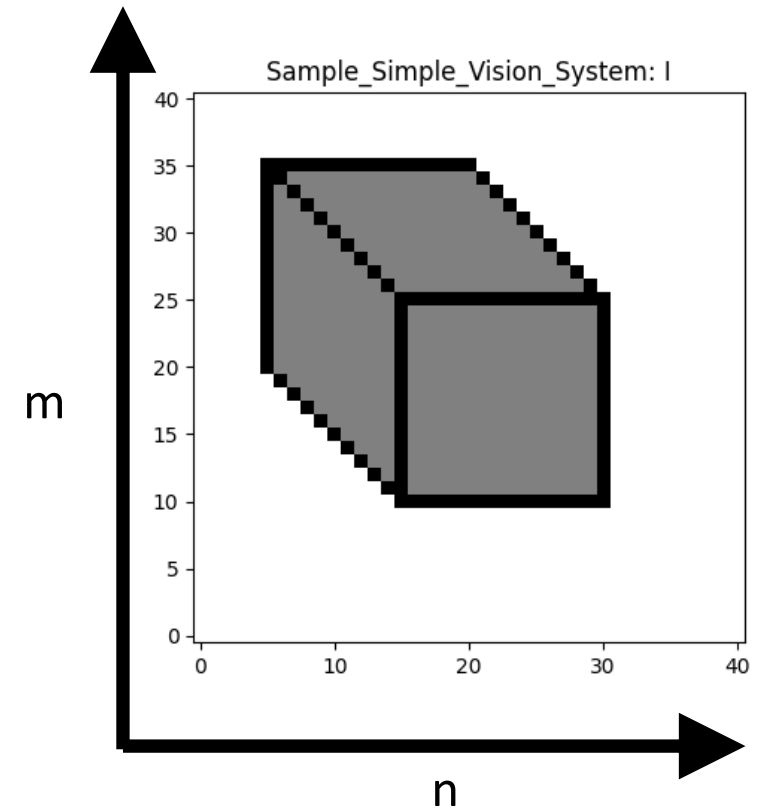
- Please make sure you have read the Lectures 4 & 5
- Please read Textbook chapter 38, at least the portions covered by the lectures.
- The notations in this slide deck may be a bit different from the lecture slides!
The goal is to make implementation easier.

Caution

- Please do not import packages (like scikit learn) that are not listed in the provided code.
- In this homework, you are NOT allowed to use NumPy's or other Python libraries' built-in **convolution, filter functions, down-sampling, up-sampling, Gaussian pyramid, and Laplacian pyramid functions**. If you use them, you will get 0 points for the entire assignment.

Convention

- In this homework, given a map (or a matrix), say I
 - $I[n, m]$ means the i -th horizontal index (left-right) and j -th vertical index (bottom-up)
 - $n \geq 0, m \geq 0$



Color images

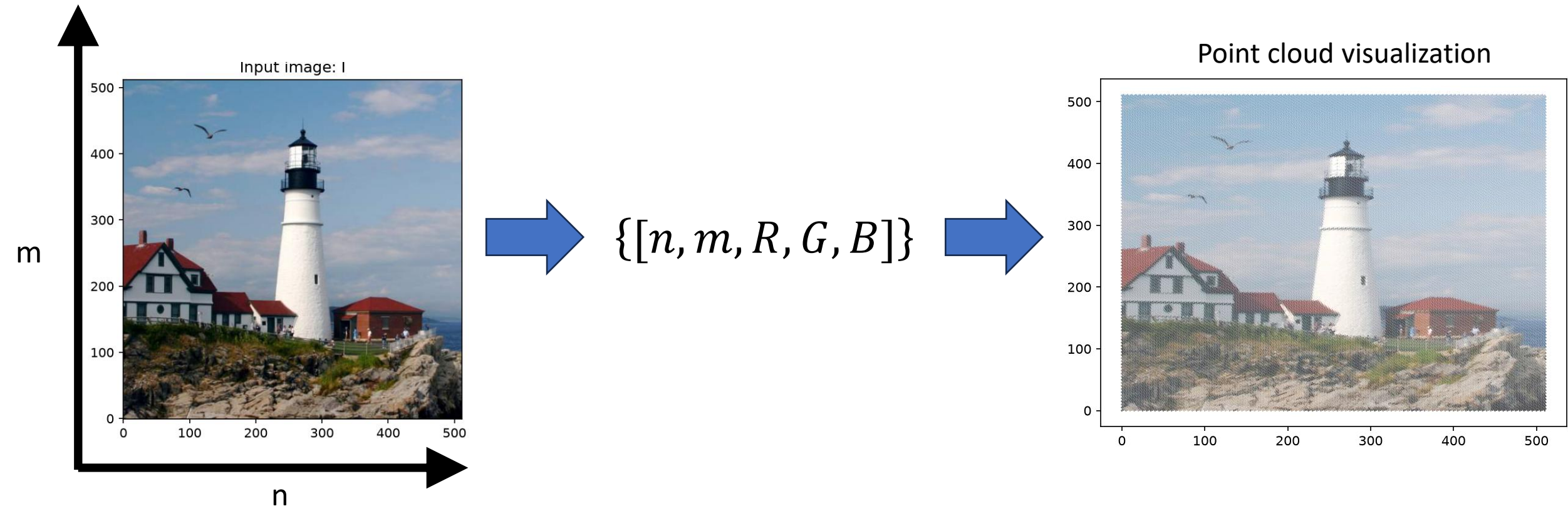
- Please note that a color image I means that the image I is a 3D tensor. The 3rd dimension corresponds to R, G, and B.



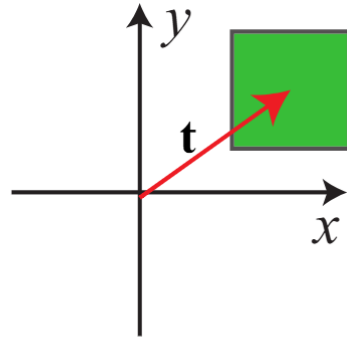
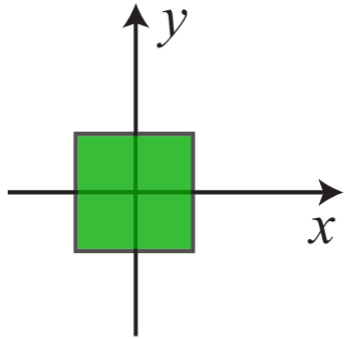
- In this homework, you will **process each channel separately**. You can extract each by $I[:, :, c]$, where c is between 0 and 2.

Question 1

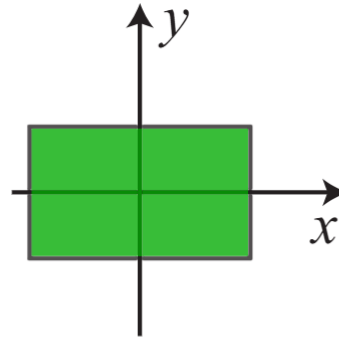
- Represent an image by a list of points and their pixel colors
 - Given an image I , where pixel coordinates start from $(0, 0)$
 - Convert it into a list of (n, m, R, G, B) , where R, G, B are associated with pixels (n, m)



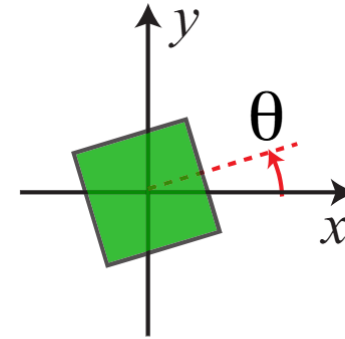
Questions 2 & 3



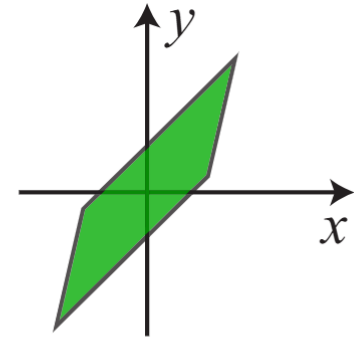
Translation



Scaling



Rotation



Shear

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & q_x & 0 \\ q_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

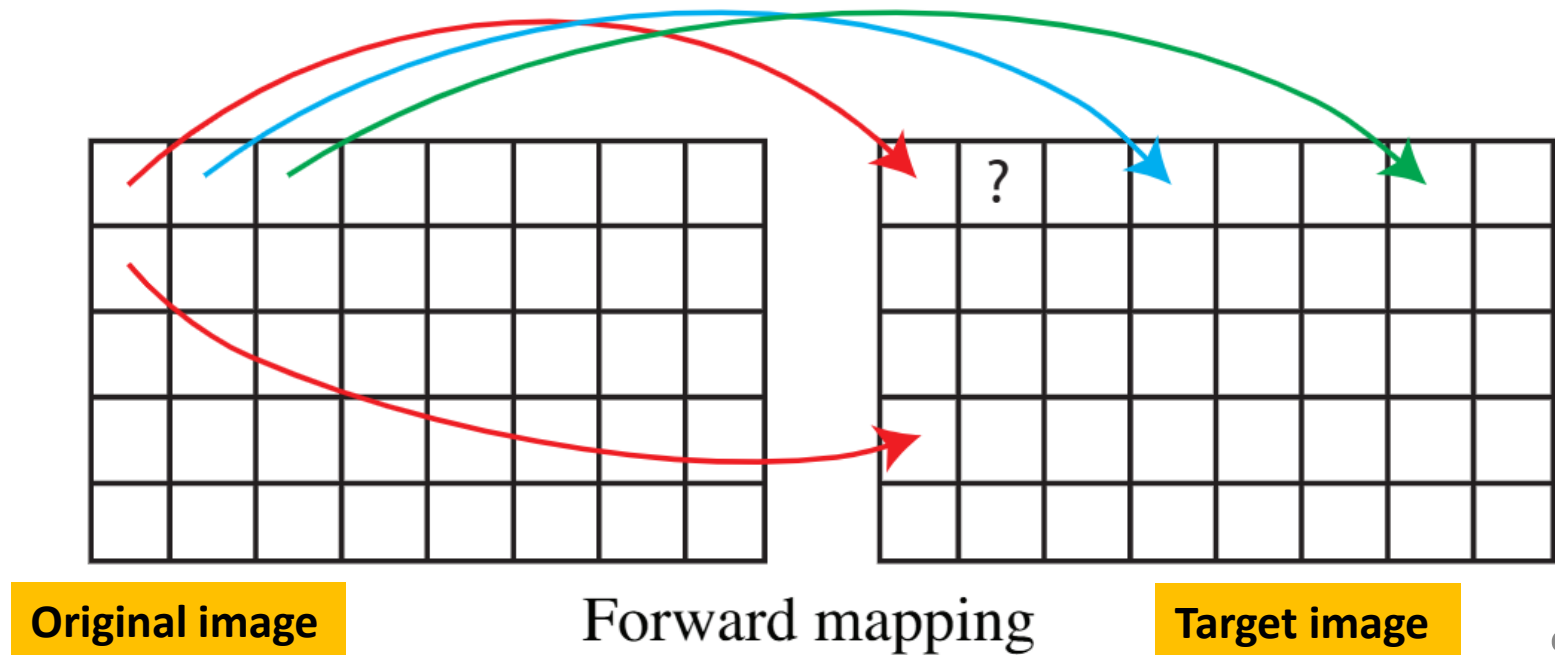
$$\mathbf{p}' = \begin{bmatrix} 1 & q_x & 0 \\ q_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \mathbf{p}$$

Questions 2 & 3

- $p' = Mp$
- Question 2:
 - (1) translate each pixel to the right by 2; to the top by 5
 - (2) rotate clockwise by 10 degrees
 - (3) scale the horizontal direction by 0.8; the vertical direction by 1.2
- Question 3:
 - (1) rotate the image counterclockwise *"w.r.t. the coordinate (250, 250)"* by 45 degrees
 - (2) scale the horizontal direction by 1.6 and the vertical direction by 1.2 *"w.r.t. the coordinate (250, 250)"*

Question 4: Forward mapping

- For every original pixel $\mathbf{p} = [n, m, 1]^T$
 - Apply the transformation $\mathbf{M}$ to the coordinate (i.e., $\mathbf{Mp}$), to map the original image to the target image
 - Round the resulting coordinate to the closest pixel coordinate (on regular spatial grid)



Question 5: Backward mapping

- For every target pixel $\mathbf{p} = [n, m, 1]^T$
 - Apply the inverse transformation $\mathbf{M}^{-1}$ to the coordinate (i.e., $\mathbf{M}^{-1}\mathbf{p}$), to map the target image back to the original image
 - Copy the color of the closest pixel coordinate (on regular spatial grid) in the original image to the target image

